🌻 Assessing quality or robustness of evidence for a causal link based on a bundle of coterminal causal claims

11 Dec 2025

This post gives more details on one of the key moments for Quality Assurance in causal mapping set out here.

Causal mapping is a way of analysing qualitative data, what people say in interviews, focus groups, reports or any written source, when you want to understand what they think causes what. This post is about a problem that surfaces once you have one of these maps in front of you. Different sources, or different parts of a single source, often make similar causal claims from the same X to the same Y, sometimes reinforcing each other, sometimes pulling in opposite directions. We call that group a *bundle* of links. How confident should we be, across the bundle as a whole, that X really does influence Y? Up to now we have largely left that judgement to the analyst's eye. Here we sketch a more systematic way to record it.

I was inspired by a recent talk by Jewlya Lynn about a causal mapping evaluation she and her team conducted Lynn (2025), in particular how they made evaluative judgments about the overall strength or robustness of evidence for the claim that one thing influenced another. We have been working on a similar idea and Jewlya's excellent report has encouraged us to move it forward.

So what is the problem exactly?

After your initial causal mapping, you will usually end up with multiple causal claims for a given single link or path from X to Y.

The Causal Map app partly grew out of our causal mapping work with QuIP evaluations. QuIP has its own way of dealing with quality and robustness of evidence. And it works mostly with relatively homogenous data sets (similar interviews with sets of similar respondents) so "number of links" can be a ballpark proxy for "strength of connection". But when working with heterogeneous sources of evidence, this does not work. In the past we have said that it is up to the analyst to look at the claims and make their own assessment about the strength of a claim that one thing influenced another, perhaps via multiple steps. But this is quite a big ask for the analyst to look at all the information from all the causal claims each time.

Just a reminder about our terminology around "links:"

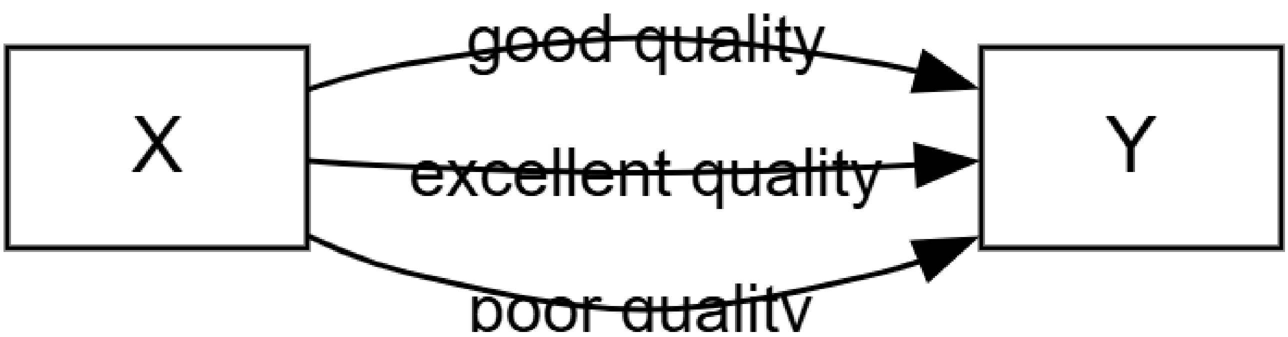
Relevant page: Bundle of Links — definition ▸

Normally, not one of these possibly hundreds or thousands of causal claims, grouped into many bundles, is incontrovertible. Sometimes we call each of these claims "evidence" but only in a weak sense of "something we could take into consideration when weighing up the validity of the claim that X causally influenced Y". Usually these links — singly or as part of bundles — have not yet passed any test at this stage or been compared to any standard.

So how can we submit our links and bundles of links to this kind of assessment?

What we can already do using existing Causal Map functionality is filter individual links. For example, we can (during or after initial coding) just add a tag `doubtful` for doubtful claims and exclude these from most or all visualisations using a "include tags" or "exclude tags" filter. But these judgements are based only on individual links. In actual evaluation and research practice there is a strong case for making more global assessments about the quality or robustness of the complete set of links within the bundle. So we add a new layer to the workflow and transform a map consisting of very many individual links within bundles into a map consisting just of a single links representing each bundle, each of which carries a global assessment of the quality or the business of the evidence within the original, unassessed, bundle.

The workflow goes from something like this (3 unassessed links):



... to something like this (1 assessed link) ....

So all four links are present in the database, but we always show either the assessed links or the unassessed links.

In more detail:

1. Analyst finishes coding the map (whether human coding, AI coding or some combination)
2. Analyst considers each bundle of links (including bundles that might contain only one link) and judges the quality/strength of the evidence
3. Analyst collapses the information in the bundle into a single new link with one or more quality assessments. In the simplest case this could be:
    1. Robust yes/no
    2. or: Robustness 1-5
4. This means that for each bundle, there is a set of usually multiple "unassessed" links and a single "assessed" link. So from now on we will normally want to view either only the assessed links or (less often) only the unassessed links.
5. Analyst uses the existing links table, pivot tables and/or map formatting to display these assessed links. Most obviously, they can show overall maps with only the quality-assessed links, formatted according to quality.

It's about saying

> I the analyst have looked at this chunk of quotes and contexts for this one bundle and I vouch for the judgement that it's enough to say yes there's something real going on here.

Unassessed view: Links table. Many links in each bundle.

Cause	Effect	Bundle	Source Count	Citation Count	Quote	Source
Ability to buy food	Food consumption quantity	Ability to buy food >> Food consu	1	1	In relation to last year, my family a	MNY-2
Farm production	Food consumption quantity	Farm production >> Food consum	7	7	In contrast to last year we barely a	MNX-6
Farm production	Food consumption quantity	Farm production >> Food consum	7	7	[Food consumption increased beca	MNX-4
Farm production	Food consumption quantity	Farm production >> Food consum	7	7	[Food consumption increased beca	MSX-1
Farm production	Food consumption quantity	Farm production >> Food consum	7	7	THE MAIN REASON OF THIS CHAN	MSY-1
Farm production	Food consumption quantity	Farm production >> Food consum	7	7	[Food consumption increased beca	MNX-3
Farm production	Food consumption quantity	Farm production >> Food consum	7	7	There has been changes as we nov	MNY-3
Farm production	Food consumption quantity	Farm production >> Food consum	7	7	IN RELATION TO THE YEAR PAST T	MSX-2
Income	Ability to buy food	Income >> Ability to buy food	1	2	In relation to last year, my family a	MNY-2
Income	Ability to buy food	Income >> Ability to buy food	1	2	In comparison to last year, we use	MNY-2
Increased knowledge	Farm production	Increased knowledge >> Farm pro	7	13	In comparison to last year where t	MNX-6
Increased knowledge	Farm production	Increased knowledge >> Farm pro	7	13	In comparison to a couple years b	MNX-3
Increased knowledge	Farm production	Increased knowledge >> Farm pro	7	13	THE POST FORMATION GIVEN BY	MSX-1
Increased knowledge	Farm production	Increased knowledge >> Farm pro	7	13	IN COMPARISON WITH THE YEAR	MSX-2
Increased knowledge	Farm production	Increased knowledge >> Farm pro	7	13	[Variety improved because...] The r	MNX-4
Increased knowledge	Farm production	Increased knowledge >> Farm pro	7	13	[Food consumption increased beca	MNX-4
Increased knowledge	Farm production	Increased knowledge >> Farm pro	7	13	In contrast to a year ago, I faced a	MNX-4
Increased knowledge	Farm production	Increased knowledge >> Farm pro	7	13	I earn money through my producti	MNY-3
Increased knowledge	Farm production	Increased knowledge >> Farm pro	7	13	[Wellbeing improved because...] Th	MNX-4
Increased knowledge	Farm production	Increased knowledge >> Farm pro	7	13	BECAUSE OF THE RECOMMENDAT	MSY-1
Increased knowledge	Farm production	Increased knowledge >> Farm pro	7	13	In relation to last year, we ate badl	MNX-3
Increased knowledge	Farm production	Increased knowledge >> Farm pro	7	13	The balanced diet of my children h	MSX-2
Increased knowledge	Farm production	Increased knowledge >> Farm pro	7	13	WITH THE GREATER PRODUCTION	MSX-1
Increased knowledge	Food consumption quantity	Increased knowledge >> Food cor	2	2	As a result of the training held in n	TWX-1
Increased knowledge	Food consumption quantity	Increased knowledge >> Food cor	2	2	Our eating habit has changed, we	TWX-2
Increased knowledge	Income	Increased knowledge >> Income	1	1	In comparison to last year, we use	MNY-2
Increased knowledge	Planted new crop/vegetabl	Increased knowledge >> Planted r	3	6	Last year we had difficulties in gett	MNX-5
Increased knowledge	Planted new crop/vegetabl	Increased knowledge >> Planted r	3	6	In comparison to last year, I used t	MNX-5
Increased knowledge	Planted new crop/vegetabl	Increased knowledge >> Planted r	3	6	In comparison to last year my proc	MNY-5
Increased knowledge	Planted new crop/vegetabl	Increased knowledge >> Planted r	3	6	[Food consumption increased beca	MNY-5
Increased knowledge	Planted new crop/vegetabl	Increased knowledge >> Planted r	3	6	[Variety increased because...] The r	MNY-5
Increased knowledge	Planted new crop/vegetabl	Increased knowledge >> Planted r	3	6	son for this is the fact that we now	MSY-3
Planted new crop/vegetabl	Food consumption quantity	Planted new crop/vegetable variet	3	3	Last year we had difficulties in gett	MNX-5
Planted new crop/vegetabl	Food consumption quantity	Planted new crop/vegetable variet	3	3	The main reason for this is the fact	MSY-3
Planted new crop/vegetabl	Food consumption quantity	Planted new crop/vegetable variet	3	3	[Food consumption increased beca	MNY-5

*Bookmark #1485 2026-04-29 07:41_*

Assessed view: Links table. One link for each unassessed bundle. Only 6 links in total.

Cause	Effect	Bundle	Source Count	Citation Count	Quote	Source
Farm production	Food consumption quantity	Farm production >> Food consum	7	7		—
Farm production	Food consumption quantity	Farm production >> Food consum	7	7		—
Planted new crop/vegetabl	Food consumption quantity	Planted new crop/vegetable variet	3	3		—
Income	Ability to buy food	Income >> Ability to buy food	1	2		—
Ability to buy food	Food consumption quantity	Ability to buy food >> Food consu	1	1		—
Ability to buy food	Food consumption quantity	Ability to buy food >> Food consu	1	1		—

*Bookmark #1484 2026-04-29 07:38_*

The big question is of course, what criteria should we use to make these robustness or quality assessments. The answer has to be based on use: what are we actually trying to do here? We might for example want to focus on the quantity or quality or robustness of the evidence taken as a whole.

In Lynn (2025), here is the matrix which Jewlya Lynn and colleagues used.

Figure C1: Strength of evidence rubric applied to determine which evidence to include

1	2	3	4	5
No evidence is available to corroborate a specific claim of a causal connection between systems dynamics, the interventions, and observed outcomes.	Evidence from a single source supports this claim	Evidence from multiple sources supports this claim (e.g., more than one document* or more than one interview). However, all sources come from similar perspectives likely to hold similar biases about how change happened. Also includes a single discussion group.	Evidence from multiple sources supports this claim. The multiple sources come from distinctly different perspectives, unlikely to hold similar biases about how change happened.	Evidence from multiple sources supports this claim, at least one of which was a discussion group. The multiple sources come from distinctly different perspectives, unlikely to hold similar biases about how change happened.
Not included	Included only as supporting evidence where aligns with other findings	Included in the analysis; if conflicting evidence was present, both claims are included.		Included and centered in the analysis; if conflicting and Level 3 or 4 evidence was present, this claim is centered.

Lynn (2025)

My feeling is that, like Jewlya, we probably want to collapse all this information to just a single dimension, perhaps 1-5.

Noting absences

Normally you'd add one assessed link for each bundle of Unassessed links, and in the UI you have a switch or filter to show either the one or the other, which then determines what you see in the outputs (maps and tables).

But there is nothing to stop you putting in an Assessed link for a bundle for which there are no claims at all. That is something which has been hard in Causal Map up to now.

Something else to think about

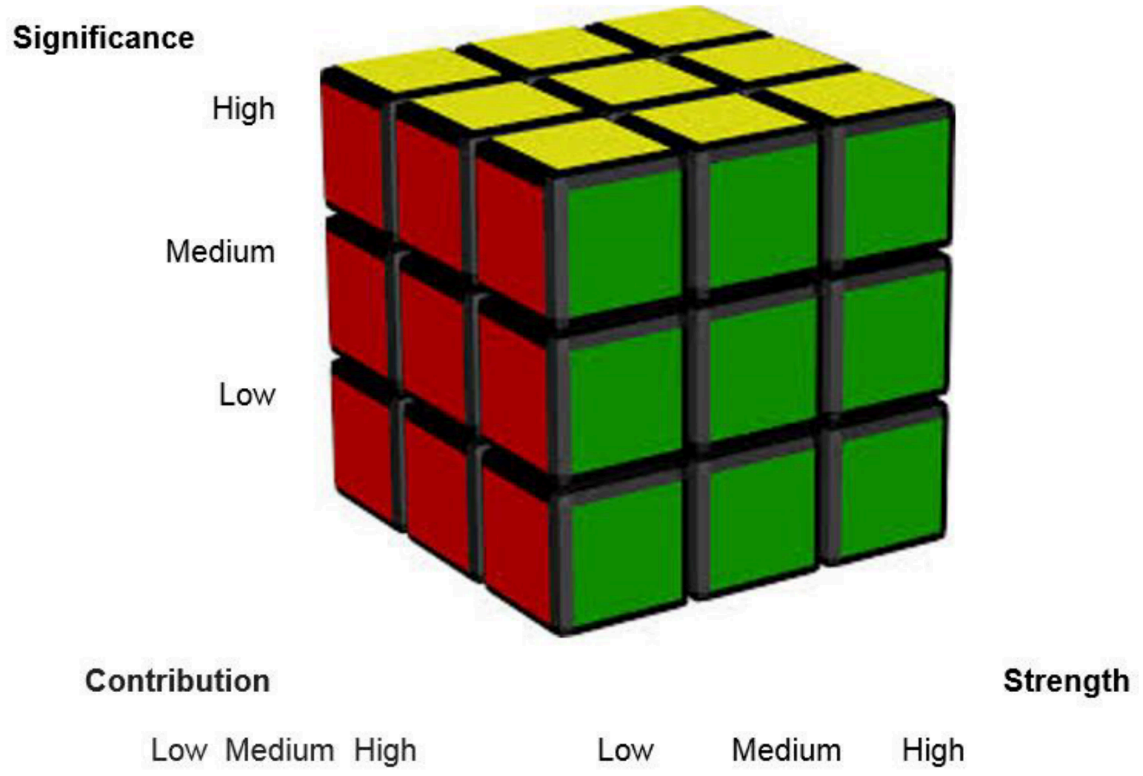
We add these "assessed" links on a per-bundle basis. But the bundles might not consist of raw labels. We can add them also for filtered labels, e.g. after zooming or after applying soft recoding.

Not so relevant in our case:

In Aston (2019), based on Mayne (2014), Tom Aston came up with this cube, which suggests an assessment based on significance, contribution and strength of evidence. This does not make quite as much sense in our context, because he is coming from a different place. For example, you are perhaps doing some kind of process tracing and are looking at one particular part of one pathway and asking what's the evidence that this foreground link is having a significant contribution to some outcome. Whereas in causal mapping you have the ambition to have captured information about all the different important contributions in parallel.

What we are proposing here is not specifically about ranking different contributions one against the other, though you could do that too.

Figure 1: Contribution Cube



Aston

(2019). Note "Strength" means "Strength of evidence".

Also Tom & Marina have a more sophisticated set of rubrics. @

References

Aston (2019). *Contribution Rubrics*. https://media.licdn.com/dms/document/media/C4D1FAQE1laRi0vrFrQ/feedshare-document-pdf-analyzed/0/1620553059516?e=1687996800&v=beta&t=bFr7dhpZ-slluV8ne1cERFelwINIaEzsQN8fiF_75gQ.

Lynn (2025). *HU Seafood Retrospective*. https://www.policysolve.com/resources/retrospective.

Mayne (2014). *Contribution Rubrics 1*.

